

Question	Marking Guidance	Mark	Comments
03.1	1. Use cold, buffered solution with same water potential; 2. Homogenise tissue and (then) filter; 3. Centrifuge at low speed and remove supernatant OR Centrifuge at low speed to remove/separate nuclei; 4. Centrifuge supernatant at higher speed (and pellet containing mitochondria is mixed with new supernatant);	4 (4 x AO1)	1. Accept isotonic for same water potential. 2. Accept blend OR grind for homogenise. 3. and 4. Accept solution for supernatant. 3. Accept heavier/denser organelles

Question	Marking Guidance	Mark	Comments
03.2	MP1 must be included for max marks. (For) 1. SDs do not overlap so significant difference/increase (in respiratory capacity); 2. Increase in ATP linked to increase in muscle contraction; OR Increase in ATP linked to increase in exercise capacity; (Against) 3. Only males (in investigation) OR No females (in investigation); 4. Only 11 (individuals in investigation) OR Small sample size; 5. RET may not strengthen required muscles OR Don't know which (skeletal) muscles affected OR Athletic performance not investigated; 6. Results may differ depending on age OR Only young (individuals) tested;	5 max (5 x AO3)	1. Accept 'difference/increase is not due to chance' for significant difference/increase. 1. Accept 'error bars' for SDs 5. Accept examples of athletic performance eg running. Ignore reference to 'no statistical test'

	7. No idea of exercise done by individuals before investigation;		
--	--	--	--